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SimPoly: Simulation of Polymers with machine learning force fields derived from first principles

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Polymers are a versatile class of materials with widespread industrial applications. Advanced computational tools could revolutionize their design, but the complex, multi-scale nature of polymers poses significant modeling challenges. Classical force fields often lack the accuracy and transferability that polymers demand, while quantum-chemical methods are prohibitively expensive for the large systems and long timescales involved. Here we develop a fast, scalable machine learning force field and show that, trained on quantum-chemical calculations alone, without any experimental data, it predicts measured bulk properties across a broad range of polymers. It reproduces polymer densities as accurately as, or better than, established classical force fields and captures the glass transition signature in the density-temperature curve, enabling the prediction of glass transition temperatures. To accelerate progress in this domain, we release a benchmark of experimental bulk properties for 130 polymers and an accompanying quantum-chemical dataset, laying the foundation for a fully in silico design pipeline for next-generation polymeric materials.

Polymeric materials are foundational to modern life, embedded in everything from the clothes we wear and the food we consume to high-performance materials in aerospace, electronics, and medicine. The ubiquity of polymers presents a significant opportunity for impact through improved tools for their functional design and a deeper understanding of their environmental and health effects. The behavior of polymeric systems is complex, spanning multiple length and time scales. This behavior arises from a combination of diverse local interactions within monomer structures and long-range interactions between polymer chains, which together determine bulk properties and processing behaviors. These complexities demand specialized modeling strategies that can capture both the molecular detail and macroscopic behavior of polymer systems[1].

Computationally efficient classical force fields (FFs) suitable for large-scale molecular dynamics (MD) simulations are currently the backbone of molecular modeling of polymers[1]. These FFs describe interatomic interactions via a series of parametrized energy terms designed to capture covalent and non-covalent (e.g., van der Waals, electrostatic) interactions. Through a time-consuming process, experts fit the parameters of these models to a combination of computational and experimental data[2–4]. A critical drawback of classical FFs is their limited transferability, i.e., their inability to accurately simulate conditions beyond those for which they were optimized. While classical FFs are successful in specialized domains with limited chemical diversity (e.g., proteins in aqueous solution), the vast chemical space of synthetic polymers limits their applicability. In addition, they are

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unable to model chemical reactions, as they rely on fixed bonding topologies. For instance, the synthesis and degradation of polymers involve complex chemical transformations that cannot be captured by classical FFs. Reactive FFs, such as ReaxFF[5], have been developed to address this limitation; however, they often require laborious reparameterization[6] as their accuracy is highly system-dependent[7].

Machine learning force fields (MLFFs) are trained on quantum-chemical data to predict energies and related quantities at a fraction of the computational cost[8]. Their applications range from high-pressure systems[9] to drug design and the discovery of molecular reaction mechanisms. MLFFs have the potential to revolutionize atomistic simulations of polymers by overcoming the fundamental limitations of classical FFs. First, so-called universal or foundational MLFFs have demonstrated transferability across diverse chemical systems[10,11]. Second, they can inherently model bond-breaking and bond-forming transformations as they do not enforce a fixed bond topology. Finally, unlike classical FFs, MLFFs can be systematically improved with more training data to accurately describe the complex interactions in the vast chemical space of polymers.

Despite their promise, progress in MLFF development has so far been driven by benchmarks focused on reproducing quantum-chemical reference data rather than experimental validation[12,13]. However, recent studies have shown that strong performance on computational benchmarks does not always translate to reliable predictions of experimental outcomes[14]. This highlights the need for robust validation against experimental data.

MLFF development has largely focused on crystalline and small-molecule benchmarks, leaving polymer systems relatively underexplored[15]. The polymer-focused work that does exist has been polymer-specific. Mohanty et al. trained an MLFF on clusters of ethylene glycol monomers, dimers, and trimers labeled with an approximate density functional, and showed that the resulting model generalized to polyethylene glycol chains of up to 100 repeat units[16]. Using system-specific quantum-chemical data, Matsumura et al. reproduced the bulk properties of individual polymers[17]. In both cases, however, the model is retrained for each polymer, and the ability to describe multiple polymers simultaneously or generalize to new ones was not assessed. Concurrent work by Xiang et al. takes a different route, learning from ab initio data of small alkanes to reproduce the density and glass transition behavior of the derived polyolefins[18].

This work makes several contributions to advance the atomistic modeling of polymers with MLFFs (Fig. 1). First, we develop Vivace, a strictly local MLFF architecture tailored for the large-scale atomistic simulations required for polymer modeling. Second, we introduce PolyArena, a benchmark for evaluating MLFFs on experimentally measured polymer properties, namely (volumetric mass) densities and glass transition temperatures. Third, we present PolyData, an accompanying dataset specifically designed for training MLFFs on these systems. Through MD simulations driven by Vivace, we accurately predict the densities of polymers, outperforming two classical FFs and matching or exceeding the performance of two existing MLFFs. Finally, we demonstrate that Vivace can capture the glass transition signature in the density-temperature curve, enabling the estimation of polymer glass transition temperatures.

Results

Experimental benchmarks and computational data for polymer simulations

To accelerate the adoption of MLFFs in polymer science, relevant and robust benchmarks grounded in experimental validation are needed. Current MLFF benchmarks primarily focus on reproducing quantum-chemical reference data, such as energies and forces, for small molecules or crystalline materials[12,13]. To this end, we introduce PolyArena, a benchmark for evaluating MLFFs on experimentally measured bulk properties of polymers. Compared to benchmarks centered around small molecules, the vast chemical diversity spanned by polymers' monomeric building blocks and their complex intra- and intermolecular interactions present unique challenges. This complexity, together with the breadth of available experimental data and their industrial importance, makes polymers an ideal test bed for developing and validating MLFFs.

PolyArena contains experimental densities and glass transition temperatures under standard conditions for 130 polymers from the Bicerano handbook[19]. These two quantities are fundamental polymer descriptors that determine a polymer's bulk characteristics, such as mechanical properties and thermal stability, and ultimately affect processing and application design[20–24]. Predicting these properties through atomistic simulation requires an accurate description of the underlying intra- and intermolecular interactions[20–24], and the glass transition in particular remains a long-standing modeling challenge[25].

Importantly, PolyArena spans polymers containing main-group elements from the first three periods, specifically H, C, N, O, F, Si, S, and Cl (Fig. 2a). A variety of polymer families are represented, among them polyolefins, polyesters, polyethers, polyacrylates, polycarbonates, polyimides, polystyrenes, siloxanes, and perfluorinated polymers. Decorating functional groups include alkyl chains, nitriles, carboxylic acid derivatives, and halogens (Fig. 2k, l). Initiating and terminating groups are assigned to each polymer considering only structural compatibility, not synthetic feasibility, to reflect the dominance of the polymer body in experimental measurements. The molecular weight of the repeating units covers a broad interval from 28 to 593 g mol^{-1}, encompassing both the simplest polyolefins and complex AB-alternating copolymers (Fig. 2b). Experimental densities under standard conditions range from 0.8 to 2.0 g cm^{-1} (Fig. 2c), and glass

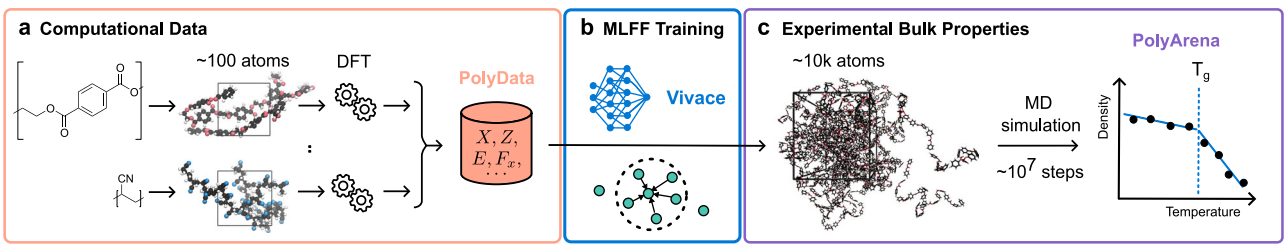


Fig. 1 | Prediction of experimental bulk properties for a wide range of polymers through MD simulations with an MLFF trained solely on ab initio data. a We generate PolyData, a targeted quantum-chemical dataset containing small, atomistic polymeric model systems that cover the complex intra- and intermolecular interactions characteristic of polymers. **b** We introduce Vivace, a fast and scalable SE(3)-equivariant MLFF architecture optimized for large-scale MD simulations, and train it on PolyData, together with a collection of other public data sets. **c** We run MD simulations driven by Vivace using large model systems of polymers to predict experimentally measured (volumetric mass) densities. We also resolve the change in slope of the density-temperature curve, enabling the determination of glass transition temperatures. To systematically evaluate the performance of MLFFs on this challenging task, we introduce PolyArena, a benchmark of experimental bulk properties for 130 polymers.

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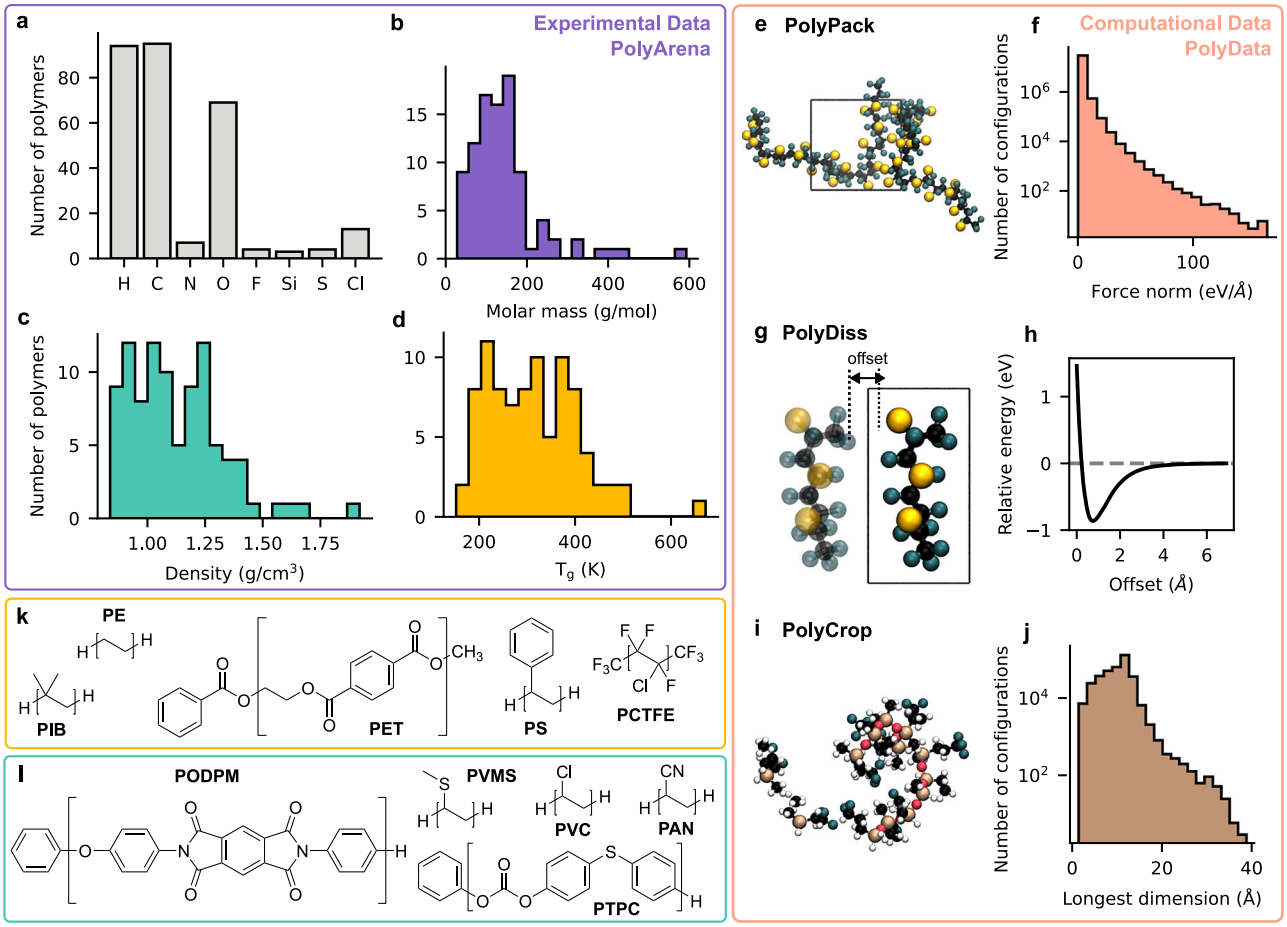


Fig. 2 | Overview of the experimental and computational data presented in this study. PolyArena (**a**–**d**) provides a benchmark for developing and systematically evaluating MLFFs on soft matter systems containing experimentally measured densities and glass transition temperatures. PolyData (**e**–**j**) is a collection of training datasets for polymeric systems, consisting of three subsets: PolyPack, PolyDiss, and PolyCrop. **a** Number of polymers containing at least one atom of the respective element. **b** Distribution of molecular weights of repeating units. **c** Distribution of experimental (volumetric mass) densities under standard conditions. **d** Distribution of experimental glass transition temperatures T_g. **e** Example structure from PolyPack, which contains structures with multiple polymer chains packed in periodic boxes at different densities. **f** Distribution of the norm of the nuclear forces in PolyPack. **g** Example structure from PolyDiss shown with one periodic image. The offset is the minimal interatomic distance across the periodic boundary. PolyDiss structures contain a single polymer chain in a periodic box of increasing size, probing intermolecular interactions at varying chain separations. **h** Dissociation curve of a single polymer chain in a periodic box. **i** Example structure from PolyCrop, which contains fragments of polymer chains in vacuum. **j** Distribution of the longest dimension of the PolyCrop clusters. **k**, **l** Repeating units of a selection of polymers from the PolyArena benchmark that are studied in more detail. Polymers that were part of the training data are shown in (**k**), while those that were not are shown in (**l**). End groups are chosen for structural stability, not synthetic feasibility. Source data are provided as a Source Data file.

transition temperatures span 152 to 672 K (Fig. 2d). Together, these distributions define a chemically and thermophysically diverse regime for benchmarking MLFFs on experimentally measurable polymer bulk properties.

To facilitate the application of MLFFs to polymers, we also introduce an accompanying collection of training datasets, called PolyData, which contains atomistic structures of polymers from PolyArena labeled with quantum-chemical methods. Existing MLFF datasets primarily focus on small molecules or crystalline materials and are thus not necessarily well-suited for capturing the complex interplay of interactions governing polymer behavior. PolyData is specifically designed to close this gap. It is composed of three subsets, PolyPack, PolyDiss, and PolyCrop (Fig. 2e–j). PolyPack and PolyDiss comprise atomistic polymeric structures with periodic boundary conditions. PolyPack features multiple packed chains with random structural perturbations that distort local geometries. In contrast, PolyDiss consists of single polymer chains in unit cells of varying sizes to modulate the distance between periodic images. Consequently, the two datasets are complementary as they target distinct interaction types. While PolyPack primarily probes strong intramolecular interactions due to local distortions, PolyDiss focuses on weaker intermolecular forces, probed by varying chain separation. Finally, PolyCrop contains fragments of polymer chains in vacuum. For details on the generation of these datasets and the quantum-chemical methods employed for labeling, we refer the reader to Methods.

Vivace—a fast, scalable, and accurate MLFF

Vivace is an SE(3)-equivariant graph neural network (GNN) engineered for the speed and accuracy required for large-scale atomistic polymer simulations. Following Allegro[26], it keeps the receptive field constant regardless of the number of message-passing iterations, which enables efficient multi-graphics-processing-unit (GPU) parallelization. While it draws inspiration from a variety of other successful models[26–31], we propose two key innovations to enhance computational efficiency while maintaining accuracy. First, Vivace employs computationally efficient SE(3)-equivariant operations. While crucial for accuracy, standard equivariant operations introduce a significant computational bottleneck, often involving expensive tensor products[32]. We find that

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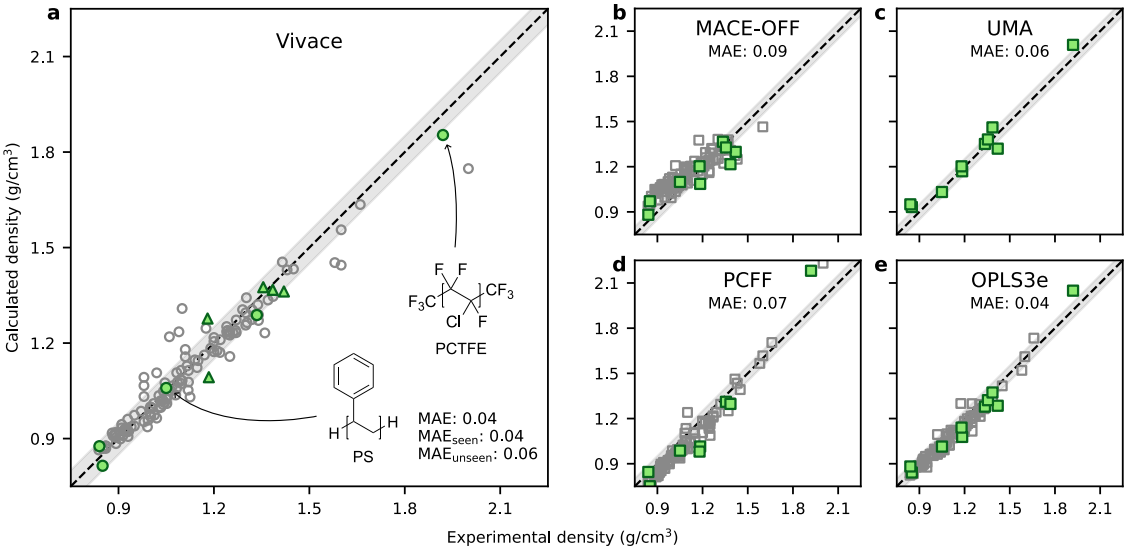


Fig. 3 | Vivace, trained solely on ab initio data, can accurately reproduce experimental densities for a wide range of polymers. In each panel, we plot calculated vs. experimental densities of polymers from the PolyArena benchmark under standard conditions for a different FF. Each marker corresponds to an MD simulation from which the density was calculated. The 10 selected polymers are indicated by green markers. Due to its slow simulation speed, UMA was only applied to these polymers. Not all polymers were simulated with the remaining models due to simulations failing (see main text for details). In **a**, circles and triangles indicate polymers seen and unseen during Vivace's training, respectively. The polymers PS (seen) and PCTFE (unseen) are annotated. In **b**–**e**, we do not make this distinction and indicate all polymers with the same square marker. MAEs (in g cm^{-3}) are reported in each panel. The dashed lines indicate perfect agreement, and the shaded areas show a deviation of ±0.05 g cm^{-3}. Source data are provided as a Source Data file.

Table 1 | Mean absolute errors (MAEs) of computed densities with respect to experimental values over 10 selected polymers and MD simulation speeds for different models

	MAE (g cm^{-3})		
Model	**Overall**	**10 polymers**	**Speed (ns d^{-1})**
Vivace	0.04	0.05	0.52
UMA[a]	0.06	0.06	0.03
MACE-OFF	0.09	0.08[b]	0.51
PCFF	0.07	0.12[c]	17.68
OPLS3e	0.04	0.06	—[d]

Simulation speeds are reported with a 0.5 fs time step and were measured with a polystyrene system containing around 8000 atoms on a single NVIDIA A100 GPU with 80 GB of memory. See Simulation Speed Measurements for details on the speed measurements.
[a]The overall MAE for UMA is based on the 10 selected polymers only.
[b]MD simulation of PCTFE failed with MACE-OFF.
[c]MD simulations of PET and PODPM failed with PCFF.
[d]We do not have access to OPLS3e to measure its speed; however, we expect it to be on the same order of magnitude as that of PCFF. MAEs are based on results from ref. 39.

learnable parameters within these operations are not essential for achieving high accuracy. Instead, Vivace captures crucial three-body interactions using a lightweight tensor product and an efficient inner-product operation. This approach is mathematically related to more complex triangular attention mechanisms[27,33] but is substantially more computationally efficient. Second, Vivace employs a multi-cutoff strategy to balance accuracy and efficiency. Efficient, invariant operations handle weaker mid-range interactions up to 6.5Å. In contrast, computationally expensive equivariant operations are reserved for short-range interactions below 3.8 Å. This strategy allows Vivace to achieve the accuracy of a large cutoff with the efficiency of a smaller one. More model details can be found in Supplementary Note 2.

Vivace was trained exclusively on computational data, with no experimental input. Specifically, Vivace was first pre-trained on a subset of OMol25[13] and PolyCrop, which contain only non-periodic structures. For subsequent fine-tuning, we introduced two datasets with periodic structures: PolyPack and PolyDiss. To retain knowledge from pre-training, Vivace was then fine-tuned on a combination of all datasets, with a higher weight assigned to the periodic configurations (see Supplementary Note 3 for details)[34]. The importance of fine-tuning on these periodic datasets is investigated in Modeling Intermolecular Interactions.

For a more detailed analysis, we selected a diverse set of 10 polymers (Fig. 2k, l). These polymers were divided into two groups of five. One group, designated as seen, was included in the training data. The other group, designated as unseen, was held out. Vivace was trained on computational data for all polymers in PolyArena except for those in the unseen group. This setup allows us to evaluate Vivace's ability to generalize to new polymers, a critical capability for a general-purpose polymer MLFF.

From ab initio data to bulk properties with MLFFs

To compute the densities of polymers in PolyArena, we perform MD simulations with atomistic model systems and analyze the resulting trajectories. We run these simulations with different FFs and compare the computed densities to experimental measurements. Concretely, we compare Vivace to two widely-used classical FFs, the Polymer Consistent Force Field (PCFF)[3] and OPLS3e[35], a refined version of the Optimized Potentials for Liquid Simulations force field, and two state-of-the-art MLFFs, MACE-OFF[36] and UMA[11]. For more details on the simulation protocol and models, we refer the reader to Methods. Not all polymers could be simulated with all models, as some simulations failed due to numerical instabilities or model incompatibilities. In Supplementary Note 9, we investigate the reasons for these failures and provide statistics on the stability of Vivace simulations. For each model, we report the mean absolute error (MAE) for all successfully simulated polymers. A separate MAE is also calculated for the 10 selected polymers. For Vivace, we further distinguish between MAEs for polymers seen and unseen during training.

Figure 3 compares computed densities to experimental densities of polymers for the different models. The results show that Vivace accurately predicts experimental densities for a wide range of polymers. For polymers seen during training, Vivace achieves an MAE of 0.04 g cm^{-3}, while for the five unseen polymers, its MAE increases to

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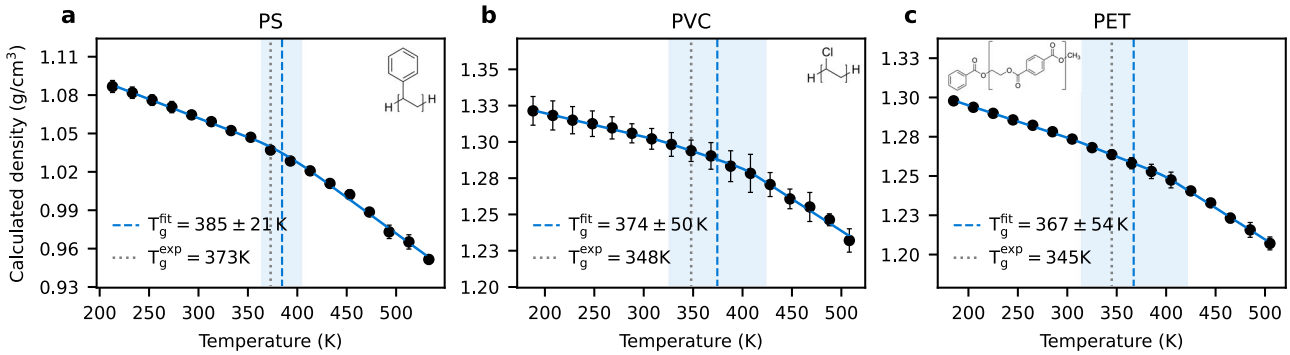


Fig. 4 | Vivace resolves the glass transition in the density-temperature curve, but the precision of the fitted transition temperature varies across polymers. Each panel shows the density-temperature curve of a different polymer, obtained with Vivace. Points and error bars give the mean density and standard deviation across three independent replicas. Blue curves show a fit of the hyperbolic function in Eq. 1 to the mean density, plotted to guide the eye. Dotted vertical lines mark the experimental T_g. Dashed vertical lines and shaded bands mark T_g^{fit} and its 68% Student-t confidence interval across three per-replica estimates (see "Methods" for details). **a** Clear change in slope, allowing precise determination of T_g^{fit}. **b** Clear change in slope, but larger per-temperature density uncertainties widen the T_g^{fit} confidence interval. **c** Less distinct transition, resulting in larger prediction uncertainty. Additional density-temperature curves are provided in the Supplementary Information. Source data are provided as a Source Data file.

0.06 g cm^{-3}. Nonetheless, the results indicate that Vivace can generalize to unseen polymers, which is a key requirement for a general-purpose polymer MLFF.

Vivace and OPLS3e achieve the lowest MAE of 0.04 g cm^{-3}, followed by UMA (0.06 g cm^{-3}), PCFF (0.07 g cm^{-3}), and MACE-OFF (0.09 g cm^{-3}). Notably, UMA and MACE-OFF reach this accuracy despite not having been trained on polymer-specific data, underscoring the transferability of these MLFFs to out-of-distribution systems. For UMA, we restricted the comparison to the 10 selected polymers due to its slow computational setup; for MACE-OFF, we had to exclude silicon-containing polymers because the model was not trained on this chemical element. Taken together, MLFFs trained solely on ab initio data match the accuracy of classical FFs whose parametrizations rely partly on experimental data, indicating that experimental measurements are not a prerequisite for accurate density predictions.

In Table 1 and Supplementary Tables 5 and 6, we compare the models on density MAE and simulation speed. Vivace and MACE-OFF are significantly faster than UMA; however, the simulation setups were not identical, so a direct comparison is difficult (see Simulation Speed Measurements for details). Nonetheless, the classical FFs remain faster than all MLFFs by at least an order of magnitude.

Observing the glass transition

The glass transition temperature, T_g, is a fundamental polymer descriptor that governs mechanical behavior, thermal stability, and processing conditions[20,21]. Predicting T_g from simulation, however, remains a long-standing challenge[25]. The glass transition is a kinetic phenomenon whose thermodynamic nature remains debated[25]; operationally, it is identified by a change in slope of the density as a function of temperature. Simulating this transition is difficult because of a timescale mismatch: experimental T_g values are measured over minutes to hours, whereas MD simulations are limited to nanosecond-to-microsecond timescales, requiring cooling rates roughly 11 orders of magnitude faster than experiment[25,37,38]. This mismatch, combined with errors in the potential energy description, causes MD-derived T_g values to often overestimate experiments by 80 to 120 K[37,39].

Here, we ask whether Vivace resolves the glass transition signature in the density-temperature diagram and whether the resulting transition temperature can be related to the experimental T_g. To this end, we simulated 29 polymers from PolyArena using 10,000-atom systems and a sequential cooling protocol: each system is held in NPT for 150 ps at each of 17 temperatures spanning $T_g \pm 160$ K in 20 K decrements (see "Methods"). We repeat the protocol with three independent replicas per polymer. The large system size, consistent with previous studies[39], is needed to reduce variance in the T_g^{fit} estimates (Supplementary Fig. 12).

The density-temperature curves generated with Vivace mostly exhibit the two linear regimes characteristic of the glass transition (Fig. 4). We fit a hyperbolic function per replica and aggregate across replicas to obtain a fitted transition temperature, T_g^{fit}, and its uncertainty (see "Methods"). Of the 29 polymers attempted, 22 yielded a fittable transition; the remaining 7 (6 seen during MLFF training and 1 not) failed via simulation instability (3 polymers) or the absence of a detectable kink in $\rho(T)$ (4 polymers; see "Methods"). To our knowledge, this is the first single MLFF, trained solely on ab initio data, to reproduce the glass transition signature in $\rho(T)$ across a chemically diverse range of polymers.

Locating the transition precisely is challenging: it manifests in $\rho(T)$ as a change in slope rather than a discontinuity, so pinpointing it requires careful uncertainty quantification[40]. Figure 4c highlights this difficulty: the change in slope is less distinct, leading to larger prediction uncertainty. For PODPM (Supplementary Fig. 5), the uncertainty even reached 86 K. Whether this stems from the simulation protocol, the force field, or the polymer itself remains an open question.

Across these 22 polymers, T_g^{fit} tends to be larger than T_g, with a mean signed deviation of +19 K and a MAE of 30 K (uncalibrated). This systematic overestimation is consistent with previous findings for classical FFs[37,39]. Despite this offset, the correlation between T_g^{fit} and experimental T_g is strong ($R^2 = 0.96$), suggesting that the transition observed in simulation and the experimental glass transition are related.

To exploit this correlation, we fit a linear function relating T_g^{fit} to experimental T_g values and use this relationship to obtain calibrated estimates, denoted T_g^{cal} (Fig. 5). The slope of the linear fit (1.20) indicates that the overestimation grows with T_g, consistent with the expected scaling of the cooling-rate-induced bias and with the empirical findings of Afzal et al.[39]. The in-sample prediction accuracy (18 K) is comparable to the in-sample calibrated value of 27 K reported by Afzal et al. for OPLS3e on a different polymer subset[39]. Refitting the calibration on the 18 polymers seen during MLFF training and evaluating on the four not seen yields a held-out MAE of 21 K.

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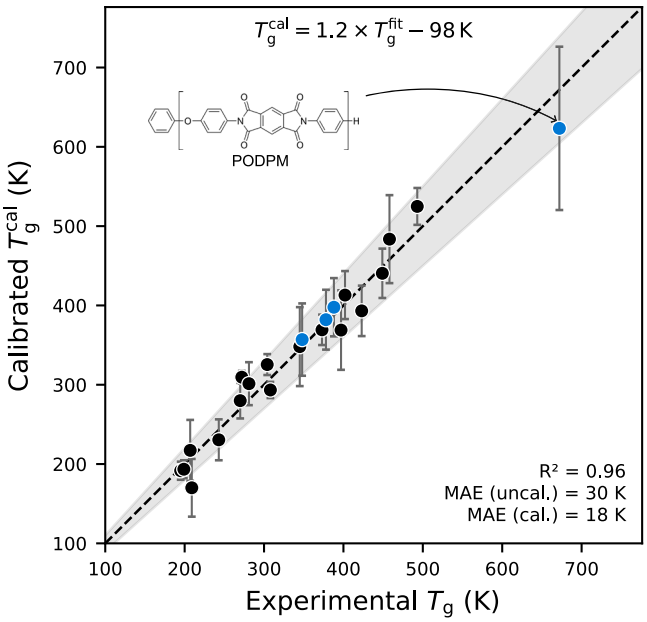


Fig. 5 | A linear calibration of the fitted transition temperatures yields glass transition temperatures in good agreement with experiment. Calibrated glass transition temperatures (T_g^{cal}) obtained with Vivace vs. experimental values (T_g) for the 22 polymers retained for analysis (the 7 polymers excluded due to simulation instability or the absence of a detectable transition are omitted). The displayed line corresponds to the linear calibration fit on all 22 polymers. Error bars indicate the 68% Student-*t* confidence interval across three per-replica estimates (see "Methods" for details). Blue markers denote polymers not seen during training. The dashed line indicates perfect agreement; the shaded area marks a 10% deviation band. Source data are provided as a Source Data file.

We probed the robustness of these results along three orthogonal axes in the Supplementary Information: the simulation protocol (a faster cooling rate and a discrete-temperature protocol with independent isothermal runs in lieu of a sequential cooling ramp, Supplementary Note 5); the degree of polymerization (varied from 10 to 50 to 200 repeating units per chain, Supplementary Note 6); and the force field (MACE-OFF and PCFF on a 10-polymer subset, Supplementary Figs. 10 and 11). We exclude UMA from this experiment due to its slow and memory-demanding simulation setup. Across all three axes the glass transition signature remains visible and reasonable T_g^{fit} predictions remain accessible. Under the faster cooling protocol, however, the kink in $\rho(T)$ became substantially harder to identify: 14 of the 29 polymers (3 simulation errors and 11 lacking a detectable kink) no longer yielded a fittable transition (Supplementary Note 5.2), highlighting the importance of the cooling rate for resolving the glass transition in MD.

Modeling intermolecular interactions

As described in Vivace—a fast, scalable, and accurate MLFF, Vivace was pre-trained on a subset of OMol25[13] and PolyCrop, both of which contain only non-periodic structures. For fine-tuning, we introduced two datasets with periodic structures: PolyPack and PolyDiss. We found that MD simulations with the pre-trained Vivace model yield densities substantially lower than their corresponding experimental values, resulting in a large density MAE of 0.60 g cm^{-3} (Supplementary Fig. 16). In contrast, fine-tuning on the periodic PolyPack and PolyDiss datasets reduces the density MAE to 0.04 g cm^{-3} (Table 2). One might hypothesize that the poor performance stems from the model not being trained on periodic data, leading to inaccurate stress tensor predictions. However, MACE-OFF, which was also not trained on periodic data, achieves a much smaller density MAE of 0.10 g cm^{-3} than the pre-trained Vivace model. The results shown in Table 2 do not support this hypothesis either. The table summarizes the performance on the PolyPack and PolyDiss test sets for the pre-trained and fine-tuned Vivace models, as well as for MACE-OFF and UMA. The fine-tuned Vivace model's stress (and force) MAE on PolyPack is comparable to that of the pre-trained model and MACE-OFF. This observation highlights how standard computational metrics, such as force and stress errors, are not always good predictors of a model's ability to run stable MD simulations[14].

The dissociation curves for the PolyDiss test set (Fig. 6) suggest an alternative explanation. The pre-trained Vivace model generally underestimates inter-chain dissociation energies, whereas the fine-tuned model more closely follows the quantum-chemical reference. We therefore hypothesize that the pre-trained model's poor MD performance stems from its inadequate description of inter-chain interactions. This hypothesis is further supported by the following observation: MACE-OFF, which predicts repulsive interactions for PODPM (Fig. 6e), yields a density that is far below the experimental value (Supplementary Fig. 10b). Together, these observations suggest that a poor description of intermolecular interactions is enough to derail bulk-property predictions.

From Fig. 6, we also see that models often underestimate the dissociation energy of polymer chains. For instance, the pre-trained Vivace model underestimates the inter-chain interactions in all cases depicted in the figure. Only after fine-tuning on PolyPack and PolyDiss does Vivace capture these interactions more accurately. Polyvinyl chloride (PVC) is a notable example, where all models predict an interaction strength significantly lower than the quantum-chemical reference method. All the models considered here are local, and we hypothesize that this is what limits them on PVC: the dipole–dipole interactions induced by the polar C–Cl bonds and the van der Waals interactions between chlorine atoms on neighboring chains decay slowly with distance, and are therefore only partially captured[41].

Effect of the cutoff radius

In an MLFF, locality is set by the cutoff radius: atoms form edges in a three-dimensional graph only when they lie within this distance of one another. A larger cutoff radius allows the model to capture interactions over greater distances. However, the number of edges scales cubically with the cutoff radius, making large cutoff radii prohibitively expensive. The cutoff radius is distinct from the receptive field, which is the maximum distance over which information can propagate through multiple message-passing iterations. It has been shown that a sufficiently large graph cutoff is necessary for an accurate force field, and that a large receptive field does not compensate for a small cutoff[42,43]. Large receptive fields also lead to poor parallelization across multiple devices, which becomes relevant to large-scale simulations involving tens of thousands of atoms. Consequently, choosing the cutoff radius involves a trade-off between accuracy and computational efficiency.

We investigated the effect of the cutoff radius, especially as Vivace's receptive field is equal to its cutoff radius: a defining feature of its architecture (see Supplementary Note 2). Figure 7a shows that a cutoff radius below 6Å fails to capture the binding energy between PCTFE polymer chains, predicting too weak an attraction that vanishes prematurely. This observation on a molecular level is consistent with the density predictions from MD simulations. From Fig. 7b, we see that MD simulations using Vivace models with cutoff radii below 5.5 Å result in density predictions that are far below the experimental values. Through visual inspection of the simulation trajectories, we found that the polymer chains unfold and repel each other, causing the system's density to collapse to near-zero values. A cutoff of approximately 6.5Å seems to be required for the simulated density to approach the experimental value. This finding is consistent with the observation that MD simulations of PCTFE with MACE-OFF, which has a cutoff of 4.5 Å, suffer from a similar density collapse. By contrast, UMA has a cutoff of

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Table 2 | Standard computational metrics do not always predict performance in MD simulations

	PolyPack		PolyDiss		PolyArena
Model	**Force MAE**	**Stress MAE**	**Force MAE**	**Stress MAE**	**Density MAE**
Vivace	183	1.9	97	1.9	0.04
Vivace (pre-trained)	179	1.7	123	2.4	0.60
MACE-OFF	188	3.1	242	19.0	0.10
UMA	110	1.0	108	1.8	0.05

This table shows MAEs for nuclear forces (in meV Å) and stress components (in meV Å) on the PolyPack and PolyDiss test sets. Density MAEs (in g cm^{-3}) from MD simulations are also included for comparison. The PolyPack and PolyDiss datasets primarily probe intra- and intermolecular interactions, respectively. The Vivace model was pre-trained on a subset of the OMol25 dataset[13] and non-periodic polymer clusters, and then fine-tuned by adding the periodic PolyPack and PolyDiss datasets (see Supplementary Note 3). Results are compared to two other MLFFs, MACE-OFF[36] and UMA[11].

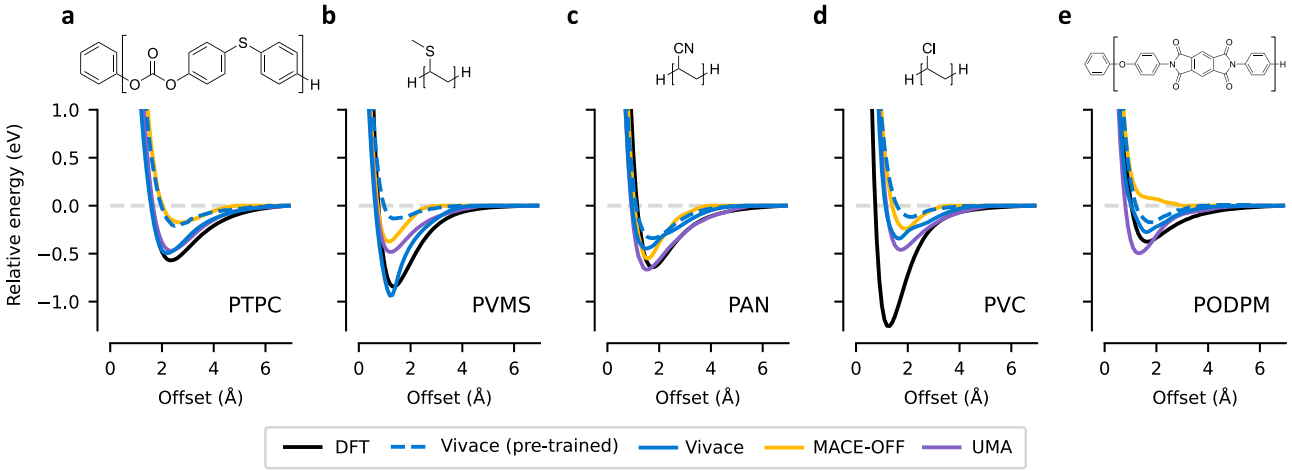


Fig. 6 | Fine-tuning Vivace improves its ability to describe inter-chain interactions. The figure shows selected dissociation curves from the PolyDiss test set, computed with four MLFFs and the density functional theory (DFT) reference method (black). Each panel plots the relative energy (in eV) as a function of the distance (offset) between polymer chains (in Å). To account for differences in absolute energies between models, the curves are shifted vertically so that the energy at the largest separation is zero. Source data are provided as a Source Data file.

6 Å and does not exhibit this issue. More recently, MACE-MH-1, a MACE foundation model with a larger 6 Å cutoff, was evaluated zero-shot on PCTFE[44]; it likewise avoids the density collapse but still underpredicts the density (1.55 g cm^{-3}, versus 1.85 g cm^{-3} for Vivace and an experimental value of 1.92 g cm^{-3}), showing that a large enough cutoff is needed to prevent the collapse but does not by itself yield an accurate density, and underscoring the value of the polymer-specific training data in PolyData. We found similar trends for other polymers, with PCTFE being particularly sensitive to this parameter, but expect this cutoff threshold to vary between polymers.

These findings validate Vivace's multi-cutoff strategy: by handling mid-range interactions within 6.5Å with efficient, invariant operations and reserving expensive equivariant operations for short-range interactions below 3.8Å, Vivace achieves the accuracy of a large cutoff with the efficiency of a smaller one.

Discussion

We demonstrated that an MLFF, trained exclusively on ab initio data, accurately predicts experimental bulk properties for a wide range of polymers. Our approach matches or outperforms traditional classical FFs, which rely on extensive, experiment-based parameterization. Furthermore, we showed that MLFFs can capture the glass transition signature in soft matter, allowing for the prediction of properties like the glass transition temperature with reasonable accuracy.

Several technical advancements made these findings possible. First, we introduced PolyArena, the first large-scale benchmark for MLFFs based on experimentally measured bulk properties of polymers. This benchmark is designed to complement existing computational ones and encourage the community to pursue more experimental validation. Second, we provide PolyData, a collection of datasets for training and fine-tuning MLFFs for polymer science applications. With broad chemical coverage, PolyData enables models to better generalize across a wide range of systems and conditions. Finally, we developed the Vivace architecture, a new, fast SE(3)-equivariant GNN optimized for efficient, large-scale simulations.

To further advance MLFFs for polymers, several challenges remain. Non-covalent long-range interactions, such as electrostatics, need to be better represented, as they are crucial in many polymers[45,46]. The MLFFs used in this work do not explicitly model these interactions, but recent progress in this area is promising[47,48]. Additionally, predicting polymer properties with MLFFs remains computationally intensive, especially for properties like T_g. For classical FFs to be replaced by data-driven alternatives in non-reactive scenarios, improvements in computational efficiency are needed. Faster simulations and a smaller memory footprint would also allow larger system sizes and longer chains, to which many properties, including the density and T_g, are sensitive.

This work highlights the transformative potential of MLFFs for polymer simulations and lays the groundwork for future research. Upcoming studies should expand this approach to other bulk properties, such as elastic moduli and thermal expansion coefficients. Most importantly, future iterations of polymer-focused MLFFs will enable applications that have been challenging for traditional methods, particularly in studying reactive processes like polymer degradation, chemical recycling, and dynamic cross-linking.

https://doi.org/10.1038/s41467-026-78736-x

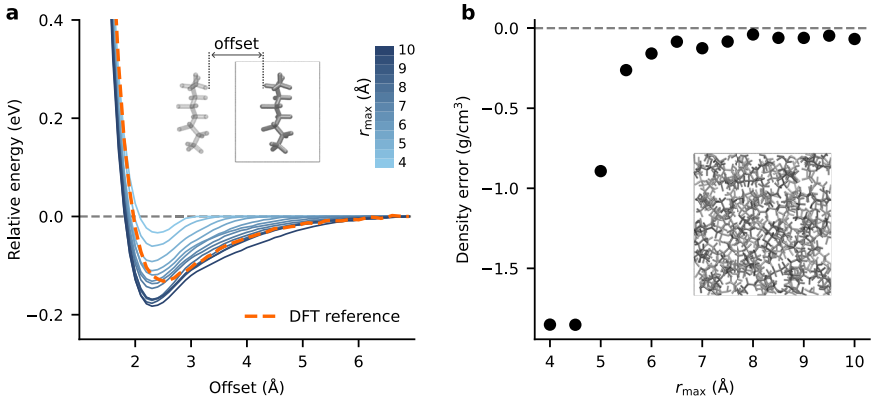


Fig. 7 | A sufficiently large cutoff radius is essential for accurate polymer simulations with MLFFs. a Dissociation curves for polychlorotrifluoroethylene (PCTFE) from the PolyDiss dataset, computed with Vivace models having different cutoff radii (r_{max}). The DFT reference is shown as a dashed orange line. The plot shows the relative energy (in eV) as a function of the distance (offset) between polymer chains (in Å). To account for differences in absolute energies between models, each curve is shifted vertically so that its energy at the largest separation is zero. The inset shows a polymer chain and one of its periodic images. **b** Error in PCTFE densities derived from MD simulations using Vivace models with varying r_{max}. A snapshot of a simulation system is shown in the inset. For cutoff radii below 5.5 Å, the polymer chains repel each other, causing the system's density to collapse. Source data are provided as a Source Data file.

Methods

Details on experimental data

Experimental measurements in PolyArena carry uncertainty from sample preparation and measurement protocol. First, the density and glass transition temperature of a given polymer sample can be affected by the conditions under which it was prepared and processed[19]. To limit the effects of sample preparation, the experimental measurements aggregated by Bicerano and employed in PolyArena represent amorphous, atactic specimens produced under standard synthesis and post-processing conditions to produce bulk form factors, avoiding those fabricated via aerosol synthesis, hot pressing, or related techniques. Second, measured T_g values depend on the heating rate, making them particularly sensitive to the measurement technique. This was observed to contribute 2 K measurement error and 7 K systematic error in the determination of the T_g for polystyrene[49]. Beyond within-method effects, reported T_g values for the same polymer can disagree by tens of degrees across literature sources[38].

The polymers in PolyArena are extracted from the Bicerano handbook[19]. We first excluded polymers with structural ambiguities or those that failed parameterization with PCFF[3] using the Enhanced Monte Carlo (EMC) program[50]. Additionally, bromine-containing polymers were removed because they were incompatible with one of the quantum-chemical basis sets employed in this study. These filters reduced the initial set of polymers by approximately 10%. While the handbook provides densities for all selected polymers, T_g values are available for only 87 of them.

Classical force field baselines

We consider two classical FFs, PCFF[3] and OPLS3e[35], as baselines. All densities computed with PCFF[3] use LAMMPS[51] as an MD engine, with parameters provided by EMC[50]. We do not run simulations with OPLS3e but use the densities and glass transition temperatures, together with associated uncertainties, reported in the Supplementary Information of the study by Afzal et al.[39]. We note that the computational setup and analysis protocol employed in that study are similar but not identical to ours, and therefore, we exercise caution when making direct comparisons.

Machine learning force field baselines

In this study, we restrict ourselves to two widely-used MLFF families, MACE-OFF23[36] and UMA[11], using the authors' released checkpoints without additional retraining or hyperparameter adaptation. MACE-OFF23[36] is a family of purely short-range MLFFs for neutral, closed-shell organic systems, built on the higher-order equivariant MACE[32] architecture, and accurately predicts a wide variety of gas- and condensed-phase properties of molecular systems. MACE-OFF23 models are trained on energies and forces using an augmented version of SPICE[52] v1 plus fragments and water clusters from QMugs[53] covering the following ten elements: H, C, N, O, F, P, S, Cl, Br, and I. There are three size variants (`s`, `m`, and `l`), and in this study, we use the fastest variant, `s`, with 64-bit floating-point precision, unless stated otherwise. For brevity, we refer to this model as MACE-OFF. Models of this family have two message-passing layers, and thus, an effective receptive field of $2 \times r_{max}$, where r_{max} is the neighbor-list cutoff radius. For the `s` variant, r_{max} = 4.5 Å.

UMA[11] is a family of so-called universal MLFFs that aim to be applicable to chemically and structurally diverse domains, such as organic molecules, molecular crystals, inorganic materials, liquids, and biomolecular fragments. Broad coverage of chemical space (more than 80 elements) is achieved by training on energies, forces, and stresses from multiple curated first-principles datasets spanning broad element combinations and bonding motifs[13]. In this work, we use the smallest variant, `UMA-s-1p1`, together with the author-provided weights most suitable for organic systems (`task_name=omol`). `UMA-s-1p1` has r_{max} = 6.0Å and four message-passing layers, resulting in a receptive field of 24.0 Å.

Simulation speed measurements

MD simulations with PCFF, Vivace, and MACE-OFF were performed using LAMMPS[51]. The MLFFs leveraged the ML-IAP interface with KOKKOS and the cuEquivariance library for fast tensor product operations. Since UMA does not have a LAMMPS interface, we use i-PI[54] as the MD engine instead. As i-PI, unlike LAMMPS, does not support multi-GPU simulations, UMA's simulation speed is only measured on a single GPU. Since i-PI is not nearly as optimized as LAMMPS, the speed measurements for UMA are not directly comparable to those of Vivace and MACE-OFF.

For speed measurements, we use the protocol detailed in MD Simulations for Density Calculations to generate polystyrene structures at a density of 1.0 g cm^{-3} with 2000, 4000, 8000, and 16,000 atoms. The generated configurations are then used in a 2.5 ps microcanonical-ensemble (NVE) simulation with a 0.5 fs time step. The simulation speeds are then reported in ns d^{-1}, representing the simulated nanoseconds per day of wall-clock time. All speed tests are

performed on a compute node with 8 NVIDIA A100(-SXM4) GPUs (80 GB each) connected via NVLink, ensuring all multi-GPU simulations run on the same node.

MD simulations for density calculations

With EMC[50] we build periodic cells containing roughly 2000 atoms at an initial density of 0.5 g cm^{-3} using chains containing 10 repeating units. Chain termini are capped with fragments resembling the repeating unit to ensure chemical stability. The generated structure is further equilibrated using a 21-step MD simulation protocol[55,56]. The aim of this protocol is to efficiently relax amorphous polymer packings to realistic densities over the course of 1.56 ns. It consists of seven cycles, each with three stages: high-temperature canonical-ensemble (NVT) simulation at T_{max}, NVT simulation at the target temperature, and isothermal-isobaric-ensemble (NPT) simulation at the target temperature with pressure ramping or release. Over the first three cycles, the pressure is ramped up to P_{max}, and over the final four cycles it is reduced to standard pressure. T_{max} is set to 700 K for all polymers to ensure it is higher than the highest T_g[56]. P_{max} must be sufficiently high to compact initially rigid configurations. Following prior work[55], we use $P_{max} = 5 \times 10^4$ bar for all systems. For more details on the 21-step protocol, including the duration of each stage, we refer the reader to Supplementary Note 8. After completing the protocol, we perform a final NPT simulation at the target temperature and standard pressure for 1 ns to compute the density.

MD simulations run with Vivace, MACE-OFF, and PCFF use the 21-step protocol detailed above with LAMMPS as the simulation engine, where a time step of 0.5 fs was used. NVT and NPT simulations used a Nosé–Hoover thermostat with a damping time of τ = 50 fs. In NPT simulations, we employed a barostat (extended Nosé–Hoover by Martyna et al.[57]) with a time constant of τ = 500 fs. We employed i-PI[54] to perform MD simulations with UMA, as a LAMMPS interface was not available. A time step of 0.5 fs was used. NVT simulations used a Langevin thermostat with a relaxation time of τ = 50 fs. NPT simulations employed the same Nosé–Hoover barostat[57] with time constant τ = 1000 fs coupled to a Langevin thermostat with relaxation time τ = 100 fs.

Predicting glass transition temperatures through simulation

We estimate T_g from MD by simulating densities along a discrete cooling protocol, fitting a hyperbolic function to the density-vs-temperature curve of each replica, and aggregating the per-replica fits. Each simulation begins from an EMC-built periodic cell containing 10,000 atoms, larger than the 2000-atom systems used for density calculations, to reduce variance in the fitted transition temperature (Supplementary Fig. 12). The cell is first equilibrated at T_g + 160 K using the 21-step protocol of MD Simulations for Density Calculations, where T_g denotes the experimental glass transition temperature of the polymer. Starting from this equilibrated configuration, we hold the system in NPT at standard pressure for 150 ps at each of 17 temperatures, decreasing the temperature by 20 K between successive stages so that the final temperature is T_g − 160 K and each stage inherits the final configuration of the previous (warmer) one. The temperature window of T_g ± 160 K brackets the transition with several stages on either side; the upper bound stays below the high-temperature regime where an MLFF, which does not enforce a fixed bonding topology, can permit bond-breaking and chain degradation, and the overall span of 320 K keeps the per-polymer simulation cost manageable while limiting the window's biasing effect on T_g^{fit}. Density statistics at each temperature are collected from the last 100 ps of the corresponding stage, with the first 50 ps discarded as equilibration. We repeat the entire protocol with three independent initial packings per polymer, a number chosen to balance statistical power against computational cost. This cooling rate is in the fast regime typical of MD-based T_g estimation, where Klajmon et al.[38] reports T_g^{fit} variations of 10–20 K across an order-of-magnitude change in cooling rate.

At each temperature, we compute the mean density of each replica from the samples in its 100 ps production window. Following Patrone et al.[58], we fit the hyperbolic function

$$\rho(T) = \rho_0 - a\left(T - T_g^*\right) - b\left[\frac{T - T_g^*}{2} + \sqrt{\frac{\left(T - T_g^*\right)^2}{4} + e^c}\right], \quad (1)$$

to each replica independently, with parameters ρ_0, a, b, c, and T_g^* to locate a transition point. We apply the same temperature window across replicas. The curve fitting is repeated over a range of initial guesses, and the fit is accepted if it lies on a stable plateau and is close to its initial guess. We define T_g^{fit} as the mean of the three per-replica transition temperatures T_g^* and report its uncertainty as a Student-t confidence interval at the 68% level (α = 0.32); this in silico quantity is distinct from the experimentally measurable glass transition temperature T_g.

The protocol does not always yield a usable T_g^{fit}, and two failure modes occur. The first is simulation instability during cooling (Supplementary Note 9): affected seeds are rerun with different random initializations, and a polymer is excluded from the analysis only if all seeds repeatedly fail (Supplementary Table 10). The second is the absence of a detectable transition in the per-replica fit. Following Patrone et al.[58], we reject fits for which the density-vs-temperature curve shows no detectable change in slope. Each rejection is confirmed by visual inspection of the density-vs-temperature curve. A polymer is excluded when no replica yields a transition.

Because T_g^{fit} systematically differs from T_g owing to the cooling-rate gap and modeling errors, empirical corrections are commonly used to bridge the two[37,39,59]. Here, we fit a linear function relating T_g^{fit} to experimental T_g across a set of polymers, yielding corrected estimates denoted T_g^{cal}.

Finally, we also analyze the topology of the polymer during these high-temperature simulations to confirm that the polymer chain stays intact (Supplementary Fig. 18 and Supplementary Note 9).

Details on computational data

The PolyData collection of datasets consists of three subsets: PolyPack, PolyDiss, and PolyCrop. To build PolyPack, we generate initial structures containing two polymer chains with EMC[50] for each polymer in PolyArena. We sample different sizes (50, 100, and 250 atoms) and a broad range of initial densities spanning 0.1× to 1.4× the experimental density to promote diversity. Each structure is subjected to a short 50 ps NVT simulation with LAMMPS[51] at 1200 K using PCFF[3]. The final frame is randomly perturbed by adding Gaussian noise with a standard deviation of 0.08Å to all atomic coordinates. This step enhances structural diversity, mitigates bias from the classical FF, and introduces high-energy configurations. However, we acknowledge that this approach may still under-sample regions of the phase space where the classical FF is inaccurate or unstable. We then perform eight geometry optimization steps, storing energies, forces, and stress tensors at every step. PolyPack contains approximately 200,000 configurations covering all polymers in the PolyArena benchmark.

PolyDiss comprises dissociation trajectories of isolated short polymer chains to probe inter-chain interactions under periodic boundary conditions. For each polymer in PolyArena, we built a polymer chain with three repeating units with EMC, placed it in an orthorhombic periodic cell, aligned it with the x-axis, and uniformly shrank the box until all inter-image atomic distances were just above 1.5 Å. Dissociation paths were then generated by incrementally

enlarging the cell in steps of 0.1 Å along *y*, *z*, *yz*, and *xyz* directions until the minimum inter-image separation reached 7.0 Å. Single-point calculations were performed for every configuration, and energies, forces, and stress tensors were recorded. PolyDiss comprises about 35,000 configurations covering all polymers in the PolyArena benchmark.

PolyCrop contains non-periodic structures and was generated using a procedure similar to that of PolyPack. The main difference is that large, periodic structures are cropped into non-periodic spherical clusters. Each cluster is centered on a randomly selected atom and includes all other atoms within a given radius, often spanning segments from multiple polymer chains. During this truncation, only C–C single bonds that cross the sphere's boundary are cleaved. Atoms outside the sphere connected by other bond types, such as double bonds, are kept to preserve the local chemical environment. Finally, all cleaved bonds are capped with hydrogen atoms. We found that this cropping process can produce highly energetic configurations. The resulting dataset contains approximately 800,000 configurations.

Quantum-chemical calculations

For the PolyPack and PolyDiss dataset, single point calculations and structure optimizations were performed with the CP2K/QUICKSTEP package[60,61], using mixed Gaussian and plane-wave basis sets and norm-conserving GTH pseudopotentials[62,63]. TZV2P basis functions were used for the expansion into the primary Gaussian basis set, and a cutoff of 1200 Ry was used for the auxiliary plane-waves expansion. The exchange-correlation energy was computed with the r^{2}SCAN functional[64] along with the D3 correction and with Becke–Johnson damping to account for dispersion interactions[65,66].

In the PolyCrop dataset, configurations were labeled with different quantum-chemical methods. Quantum mechanical calculations were performed with the ORCA program[67], using the range-separated hybrid ωB97X-V density functional[68] and the def2-TZVPP basis set[69]. All structures were treated as closed-shell, with a spin multiplicity of one. If a structure was too large to be labeled with DFT within a day, it was labeled with the semi-empirical GFN2-xTB tight-binding method[70] instead. Approximately 400,000 clusters containing up to 234 atoms were labeled with ORCA, and another 400,000 clusters with up to 1411 atoms were labeled with GFN2-xTB.

Data availability

The PolyData training data generated in this study and the Vivace model checkpoint are available on https://huggingface.co/microsoft/simpoly. The PolyArena benchmark, the raw simulation data, and the source data generated in this study are available on https://github.com/microsoft/simpoly. The experimental densities and glass transition temperatures compiled in PolyArena were taken from the handbook of Bicerano[19]. Source data are provided with this paper.

Code availability

The code that supports this study is available on https://github.com/microsoft/simpoly, and the model checkpoint that serves as a starting point for users' calculations is available on https://huggingface.co/microsoft/simpoly.

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Competing interests

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Additional information

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